

Integración (P4)

Sea $f: U \subseteq \mathbb{C} \rightarrow \mathbb{C}$, U abierto,

Prop. En $\mathbb{R}^2$, $\gamma: [a, b] \rightarrow \mathbb{R}^2$
 $t \mapsto (x(t), y(t))$

Def1: γ se dice suave si γ es C^1 y $|\gamma'(t)| \neq 0$.

Def2: $\gamma: [a, b] \rightarrow \mathbb{C}$ se dice suave a trozos si $\exists a = a_0 < a_1 < \dots < a_n = b$ y $\gamma_i: [a_i, a_{i+1}] \rightarrow \mathbb{C}$ tq $\gamma_i(a_{i+1}) = \gamma_{i+1}(a_{i+1})$ (γ continua) y γ_i suave $\forall i$.

Obs: $\gamma'(a)$, $\gamma'(b)$ derivadas laterales en los bordes. No tienen por qué coincidir cuando digo suave a trozos.

Sea $A: [a, b] \rightarrow \mathbb{C}$, $A(t)$ función con $t \in [a, b]$

$\Rightarrow A(t) = u(t) + iv(t)$ con $u, v: [a, b] \rightarrow \mathbb{R}$

Def $\int_a^b A(t) dt = \int_a^b u(t) dt + i \int_a^b v(t) dt$

Obs: u, v son integrables en $[a, b]$

Prop Sea $A: [a, b] \rightarrow \mathbb{C}$, $A = u + iv$

$$\text{I. } \frac{d}{dt} \int_a^t A(s) ds = A(t)$$

$$\text{P/ } \frac{d}{dt} \left(\int_a^t u(s) ds + i \int_a^t v(s) ds \right) \stackrel{\text{TFC en } \mathbb{R}}{=} u(t) + iv(t) = A(t)$$

$$\text{II. } A'(t) = u'(t) + i v'(t) \Rightarrow \int_a^b A'(s) ds = A(b) - A(a)$$

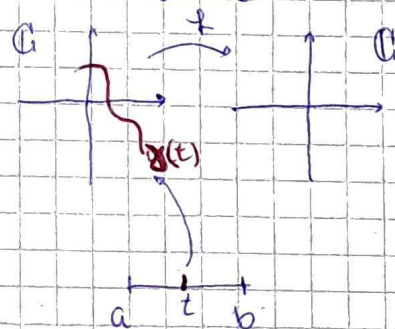
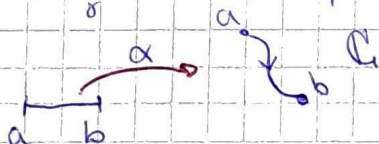
P/ Ejercicio

Sea $f: U \subseteq \mathbb{C} \rightarrow \mathbb{C}$, U ab. y $\alpha: [a, b] \rightarrow \mathbb{C}$ una curva suave

$$\Rightarrow \int_{\gamma} f(z) dz = \int_a^b f(\alpha(t)) \cdot \alpha'(t) dt \in \mathbb{C}$$

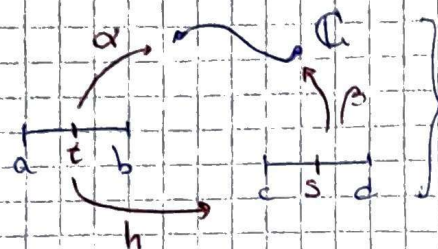
Acá, $A(t) = f(\alpha(t)) \cdot \alpha'(t): [a, b] \rightarrow \mathbb{C}$

1) En $\int_{\gamma} f(z) dz$ hay una orientación de γ



Parametrización de una curva γ , $\text{Im}(\alpha) = \gamma$

2) Si $\alpha: [a,b] \rightarrow \mathbb{C}$ y $\beta: [c,d] \rightarrow \mathbb{C}$ dos parametrizaciones suaves de la misma curva con la misma orientación y C'
 $\Rightarrow \int_a^b f(\alpha(t)) \alpha'(t) dt = \int_c^d f(\beta(s)) \beta'(s) ds$.
 la definición no depende de la parametrización siempre que respeten la orient.



$\exists h: [a,b] \rightarrow [c,d]$ tq h biyectiva, C' y $h'(t) > 0$.

$$\Rightarrow \beta(h(t)) = \alpha(t)$$

$$\beta'(h(t)) \cdot h'(t) = \alpha'(t)$$

$$\text{Luego } \int_a^b f(\alpha(t)) \alpha'(t) dt = \int_a^b \underbrace{f(\beta(h(t)))}_{\in \mathbb{C}} \underbrace{\beta'(h(t))}_{\in \mathbb{C}} \underbrace{h'(t)}_{\in \mathbb{R}} dt =$$

$$= \int_c^d f(\beta(s)) \beta'(s) ds$$

$$\uparrow \quad h(t)=s \Rightarrow h'(t)dt=ds \text{ pues } h'(t) > 0$$

¿y si γ es suave a trozos?

Def $\int_{\gamma} f(z) dz = \sum_{j=0}^{n-1} \int_{\gamma_j} f(z) dz$ con γ_j suave entre a_j y a_{j+1}

Def $\text{long}(\gamma) = \int_a^b |\gamma'(t)| dt$ es la longitud de la curva

Prop Sean $f, g: U \rightarrow \mathbb{C}$, $U \subseteq \mathbb{C}$ abierto. Sea $a \in \mathbb{C}$, entonces

I. $\int_{\gamma} (af + bg)(z) dz = a \int_{\gamma} f(z) dz + b \int_{\gamma} g(z) dz$ Integral lineal

II. $\int_{\gamma} f(z) dz = - \int_{\tilde{\gamma}} f(z) dz$ si γ y $\tilde{\gamma}$ tienen orientaciones opuestas

III. $|\int_{\gamma} f(z) dz| \leq \int_{\gamma} |f(z)| |dz|$, donde $|dz| = |\gamma'(t)| dt$.

~~$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$~~ $= \int_a^b f(\gamma(t)) |\gamma'(t)| dt$

Teo Sea $f_n: U \subseteq \mathbb{C} \rightarrow \mathbb{C}$ tq $f_n \rightarrow f$ unif. sobre compactos de U (o sea dado $K \subseteq U$ compacto y $\epsilon > 0$, $\exists N$ tq $|f_n(z) - f(z)| < \epsilon \forall z \in K \forall n \geq N$) y γ curva suave a trozos, $\gamma: [a,b] \rightarrow \mathbb{C}$. Entonces

$$\int_{\gamma} f_n(z) dz \xrightarrow{n \rightarrow \infty} \int_{\gamma} f(z) dz$$

Y γ es compacto, tomo $\frac{\epsilon}{\text{long}(\gamma)}$

$$\exists N \text{ tq } n \geq N \quad |f_n(z) - f(z)| < \frac{\epsilon}{\text{long}(\gamma)} \quad \forall z \in \gamma, \forall z \in \gamma(t), t \in [a,b]$$

Mención del jardínero.

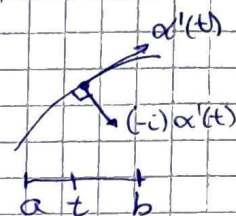
$$\left| \int_{\gamma} f_n(z) dz - \int_{\gamma} f(z) dz \right| \leq \int_{\gamma} \underbrace{|f_n(z) - f(z)|}_{< \varepsilon / \text{long}(\gamma)} |dz| < \frac{\varepsilon}{\text{long}(\gamma)} \int_{\gamma} |dz| = \varepsilon$$

Prdo (Análisis II) $\int_a^b \langle F(\alpha(t)), \alpha'(t) \rangle dt$ con α suave, $\alpha: [a, b] \rightarrow \mathbb{C}$.

Si $\alpha: [a, b] \rightarrow \mathbb{C}$ param. por long. de arco, $|\alpha'(t)| = 1$

$T(t) = \alpha'(t)$ es tangente a γ en $\alpha(t)$

$N(t) = (-i)\alpha'(t)$ es normal a γ en $\alpha(t)$



Ahora $\{T(t), N(t)\}$ es B.O.N de $\mathbb{R}^2$.

$f(z) = u + iv \Rightarrow \bar{f}(z) = u - iv$. Pensamos $\alpha(t) = (\alpha_1(t), \alpha_2(t)) = \alpha_1 + i\alpha_2$

$\bar{f}(z) = (u, -v) = \langle \bar{f}, T \rangle T + \langle \bar{f}, N \rangle N$ con el pi usual de $\mathbb{R}^2 \ni (u, -v)$

$$(u, -v) \cdot (\alpha_1', \alpha_2')$$

$$(u, -v) \cdot (\alpha_2', -\alpha_1')$$

$$u\alpha_1' - v\alpha_2'$$

$$u\alpha_2' + v\alpha_1'$$

$$= \text{Re}(f \cdot \alpha')$$

$$= \text{Im}(f \cdot \alpha')$$

$$\Rightarrow \bar{f}(z) = \text{Re}(f \cdot \alpha') T + \text{Im}(f \cdot \alpha') N$$

$$\text{Luego, } \int_{\gamma} f(z) dz = \int_a^b \langle \bar{f}, T \rangle dt + i \int_a^b \langle \bar{f}, N \rangle dt$$

$$\text{Pues } \int_{\gamma} f(z) dz =$$

$$= \int_a^b f(\alpha(t)) \alpha'(t) dt =$$

$$= \int_a^b \text{Re}(f(\alpha(t)) \alpha'(t)) dt +$$

$$+ i \int_a^b \text{Im}(f(\alpha(t)) \alpha'(t)) dt$$

$\Rightarrow \int \text{Re}(f(z) dz) =$ trabajo realizado por $\bar{f}(z)$ a lo largo de γ .

$\int \text{Im}(f(z) dz) =$ flujo de $\bar{f}(z)$ a través de γ .

Ej (♥♥♥)

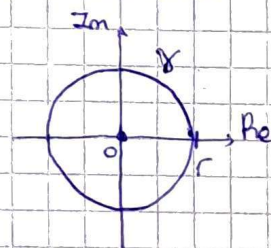
o) γ circunferencia de radio r : $t \mapsto re^{it}$ con $t \in [0, 2\pi]$ es una curva cerrada suave

..) $f(z) = z^n$ con (supong.) $n \geq 0$. ~~(n) \in \mathbb{Z}~~ ($n \in \mathbb{Z}$)

¡Calculamos la integral! $\alpha'(t) = rie^{it}$

$$\int_{\gamma} f(z) dz = \int_{\gamma} z^n dz = \int_0^{2\pi} r^n e^{int} r e^{it} i dt =$$

$$= r^{n+1} i \int_0^{2\pi} e^{i(n+1)t} dt = r^{n+1} i \left(\frac{e^{i(n+1)t}}{i(n+1)} \right) \Big|_0^{2\pi} = 0$$



Esto también vale para $n < -1$. ¿Qué pasa en $n = -1$?

$$\boxed{\int_{\gamma} \frac{1}{z} dz = \int_0^{2\pi} \frac{e^{it} i}{e^{it}} dt = 2\pi i}$$

02/09/2024

Def: $\gamma: [a, b] \rightarrow \mathbb{C}$, $\gamma = \gamma_1 + i\gamma_2$

$$\text{Def } \int_{\gamma} \gamma(t) dt = \int_a^b \gamma_1(t) dt + i \int_a^b \gamma_2(t) dt$$

$$\text{Def } \gamma \text{ suave (suave a trozos)}, f: U \rightarrow \mathbb{C}, \int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

Prop: Si $F'(z) = f(z)$, F holomorfa

$$\int_{\gamma} f(z) dz = \int_{\gamma} F'(z) dz = \int_a^b F'(\gamma(t)) \gamma'(t) dt = F(\gamma(b)) - F(\gamma(a))$$

En particular, si f tiene una primitiva entonces $\int_{\gamma} f(z) dz = 0$ para cualquier curva cerrada

$$\Rightarrow \int_{\gamma} f(z) dz = 0 \text{ con } \gamma \text{ curva cerrada}$$

2) $\int_{\gamma} f(z) dz$ solo depende de los extremos de la curva

$$\Rightarrow \int_{\gamma} f(z) dz = \int_{\theta} f(z) dz \text{ con } \gamma, \theta \text{ que tienen}$$

$$\text{los mismos extremos } \begin{cases} \gamma: z_1 \rightarrow z_2 \\ \theta: z_1 \rightarrow z_2 \end{cases} \Rightarrow \gamma = z_1 \xrightarrow{\text{orient. opuesta}} z_2 \xrightarrow{\text{orient. opuesta}} z_1$$

$$\text{luego, curva cerrada} \Rightarrow 0 = \int_{\gamma} f(z) dz = \int_{\gamma} f(z) dz - \int_{\theta} f(z) dz$$

$$\Rightarrow \int_{\gamma} f(z) dz = \int_{\theta} f(z) dz$$

Teo $\sup f: U \rightarrow \mathbb{C}$, $U \subseteq \mathbb{C}$ abierto, f continua tq $\int_{\gamma} f(z) dz = 0$ $\forall \gamma$ curva cerrada

$$\Rightarrow \exists F: U \rightarrow \mathbb{C} \text{ holomorfa tq } F'(z) = f(z) \text{ (f tiene primitiva)}$$

$\nabla z_0 \in U$ fijo. $\sup U$ conexo (¿por qué podemos suponer esto?)

$$\text{Para } z \in U, \gamma: z_0 \rightsquigarrow z \quad \gamma(0) = z_0, \gamma(1) = z$$

$$\text{Def } F(z) = \int_{\gamma} f(z) dz$$

Obs: F está bien definida pues de la hip $\int_{\gamma} f(z) dz = 0 \quad \forall \gamma$ curva cerrada

$$\Rightarrow \int_{\gamma} f(z) dz \text{ solo depende de los extremos "}$$

que de $f(z)$

$$\frac{F(z+h) - F(z)}{h} \text{ Quiero ver si existe l'ím de eso}$$

$$\text{Elijo } |h| \text{ suf. pequeño tq } |h| < r, D(z, r) \subseteq U$$



$$\gamma: z_0 \rightsquigarrow z$$

$$F(z) = \int_{\gamma} f(z) dz$$

Para $F(z+h)$ agrego "el palito" que une z , $z+h$ a la curva

$$F(z+h) = \int_{\gamma \cup \text{seg } z, z+h} f(z) dz$$

$$\Rightarrow \frac{F(z+h) - F(z)}{h} = \frac{\int_{\gamma \cup \text{seg}} f(z) dz - \int_{\gamma} f(z) dz}{h} = \int_{\text{seg } z, z+h} \frac{f(z)}{h} dz \quad (*)$$

Quedo parametrizar el segmento:

$t \in [0, 1]$, $z+th$ es un ~~camino~~ segmento que une

Así $\tilde{\gamma} = \gamma \cup \text{seg}$ me une z_0 , $z+h \Rightarrow F(z+h) = \int_{\tilde{\gamma}} f(z) dz$

$$(*) = \int_{\text{seg}} \frac{f(z)}{h} dz = \frac{1}{h} \int_{\text{seg}} f(z) dz = \frac{1}{h} \int_0^1 f(z+th) \cdot h dt = \int_0^1 f(z+th) dt$$

deb de la integral en una curva $\forall |h| < r$

Ahora:

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| = \left| \int_0^1 (f(z+th) - f(z)) dt \right| \leq \int_0^1 |f(z+th) - f(z)| dt \quad (*)$$

Dado $\varepsilon > 0$, $\exists R > 0$ tq si $|h| < R$, $|f(z+th) - f(z)| < \varepsilon \quad \forall t \in [0, 1]$ (f es continua)

$$(*) \leq \varepsilon \quad \text{si } |h| < R$$

$$\therefore \lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h} = f(z) \quad \Leftrightarrow F'(z) = f(z) \quad (F \text{ holomorfe})$$

Obs: La misma demostración funciona si uno considera curvas γ poligonales con segmentos paralelos a los ejes.

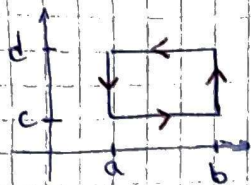
Prop Sea $U \subseteq \mathbb{C}$ abierto, $f: U \rightarrow \mathbb{C}$ continua. Son equivalentes:

I. $\int_{\gamma} f(z) dz = 0 \quad \forall \gamma$ curva cerrada

II. $\int_{\gamma} f(z) dz$ solo depende de los extremos de γ

III. f tiene primitiva: $\exists F$ holomorfe tq $F'(z) = f(z) \quad (F: U \rightarrow \mathbb{C})$

Si R es un rectángulo



∂R es curva suave a trozos recorrida en sentido antihorario.

Teo (Green a rectángulos). Sea $U \subseteq \mathbb{C}$ abierto, $f: U \rightarrow \mathbb{C}$ clase $\mathcal{C}^1$, R un rectángulo. Entonces:

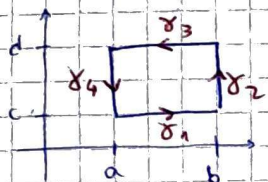
$$\iint_R \frac{\partial f}{\partial \bar{z}} dx dy = \int_{\partial R} f(z) dz \cdot \frac{1}{2i}$$

$\downarrow$
 $\mathbb{R}^2 \rightarrow \mathbb{R}^2$

Como: Si f fuera holomorfa $\Rightarrow \frac{\partial f}{\partial \bar{z}} = 0 \Rightarrow \int_{\partial R} f(z) dz = 0$.

estamos a altura d pero recorrimos de a a b y quiero de b a a
 $\int_{\gamma_3} f(z) dz - \int_{\gamma_1} f(z) dz$

D/(Teo)



$$\iint_R \frac{\partial f}{\partial y} dx dy \stackrel{\text{Fubini}}{=} \int_a^b \left(\int_c^d \frac{\partial f}{\partial y} dy \right) dx = \int_a^b (f(x, d) - f(x, c)) dx =$$

$$\iint_R \frac{\partial f}{\partial x} dx dy = \int_c^d \left(\int_a^b \frac{\partial f}{\partial x} dx \right) dy = \int_c^d (f(b, y) - f(a, y)) dy$$

$\parallel$
de la parametrización $-i \left[\int_{\gamma_2} f(z) dz + \left(- \int_{\gamma_4} f(z) dz \right) \right]$

$$\begin{aligned} \Rightarrow \iint_R \frac{\partial f}{\partial \bar{z}} dx dy &= \iint_R \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) dx dy = \frac{1}{2} \iint_R \frac{\partial f}{\partial x} dx dy + i \frac{1}{2} \iint_R \frac{\partial f}{\partial y} dx dy = \\ &= \frac{1}{2i} \int_{\partial R} f(z) dz \quad (*) \\ &= \sum_{i=1}^4 \left(\int_{\gamma_i} f(z) dz \right) \end{aligned}$$

$$\begin{aligned} (*) &= -\frac{1}{2} i \int_{\gamma_2} f(z) dz - \frac{1}{2} i \int_{\gamma_4} f(z) dz + \frac{1}{2} i \left(- \int_{\gamma_3} f(z) dz - \int_{\gamma_1} f(z) dz \right) = \\ &= -\frac{1}{2} i \left(\int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz + \int_{\gamma_3} f(z) dz + \int_{\gamma_4} f(z) dz \right) = \frac{1}{2i} \int_{\gamma} f(z) dz \quad \square \end{aligned}$$

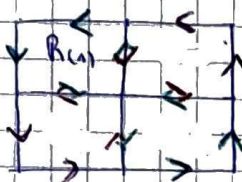
Teo Cauchy Sea $U \subseteq \mathbb{C}$ abierto, $f: U \rightarrow \mathbb{C}$ $\mathcal{C}^1$ y holomorfa. Entonces

$$\int_{\partial R} f(z) dz = 0 \quad \forall R \text{ rectángulo.}$$

Teo Cauchy-Goursat Sea $f: U \rightarrow \mathbb{C}$ holomorfa, $\int_{\partial R} f(z) dz = 0 \quad \forall R \subseteq U$

D/NOT: $\eta(R) = \int_{\partial R} f(z) dz$

$$\int_{\partial R} f(z) dz = \sum_{i=1}^4 \left(\int_{\partial R_i} f(z) dz \right) = \sum_{i=1}^4 \eta(R_i)$$



$$\sup \eta(R_i) \geq \frac{1}{4} \eta(R)$$

Ahora agamo R_{i1} y lo parto en 4



$$\text{Per}(R_{i1}) = \frac{1}{2} \text{Per}(R_i)$$

y agamo R_{i2}

$$\text{Diam}(R_{i1}) = \frac{1}{2} \text{Diam}(R_i)$$

$$\Rightarrow \eta(R_{i2}) \geq \frac{1}{4} \eta(R_{i1}) \geq \frac{1}{4^2} \eta(R)$$

Así armamos una suc. $\{R_n\}$ hem de rectángulos tq

$$\text{Per}(R_{i2}) = \frac{1}{2^2} \text{Per}(R_i), \text{diam}(R_{i2}) = \frac{1}{4} \text{diam}(R_i), \text{diam}(R_n) = \frac{1}{2^n} \text{diam}(R)$$

$$\text{Per}(R_n) = \frac{1}{2^n} \text{Per}(R)$$

$$4^n \eta(R_n) \geq \eta(R) \quad (*)$$

$$\Rightarrow \exists \{z^*\} = \bigcap_{n=1}^{\infty} R_n$$

f holomorfe

$$(*) \quad |f(z) - f(z^*) - f'(z^*)(z - z^*)| \leq \frac{\epsilon}{\text{diam}(R) \cdot \text{Per}(R)} |z - z^*|$$

Elegimos R_n tq $R_n \subseteq D(z^*, \delta)$. Sea $g(z) = 1$, tiene primitiva $G(z) = z$
 $\hat{g}(z) = z, \dots, \hat{G}(z) = \frac{z^2}{2}$

$$(*) \quad \left| \frac{f(z) - f(z^*)}{z - z^*} - f'(z^*) \right| < \frac{\epsilon}{\text{Per}(R) \cdot \text{diam}(R)}$$

$$\left| \frac{f(z) - f(z^*) - f'(z^*)(z - z^*)}{z - z^*} \right|$$

Porque son holomorfas sobre una curva cerrada.

$$(*) \quad |\eta(R)| \leq 4^n |\eta(R_n)| \quad (\text{Chequear})$$

z, z^* están en el rectángulo

$$|\eta(R_n)| = \left| \int_{\partial R_n} f(z) dz - \int_{\partial R_n} f(z^*) dz - \int_{\partial R_n} f'(z^*)(z - z^*) dz \right|$$

$$\leq \int_{\partial R_n} |f(z) - f(z^*) - f'(z^*)(z - z^*)| |dz| \leq \int_{\partial R_n} \frac{\epsilon}{\text{Per}(R) \cdot \text{diam}(R)} |z - z^*| |dz|$$

$$\leq \frac{1}{2^n} \frac{\epsilon}{\text{Per}(R)} \int_{\partial R_n} |dz| \leq \frac{1}{2^n} \frac{\epsilon}{2^n} \frac{1}{2^n}$$

$$\text{Long}(\partial R_n) = \text{Per}(R_n)$$

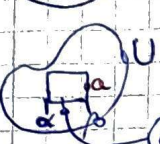
Volviendo, $| \eta(R) | \leq 4^n | \eta(R_n) | \leq 4^n \cdot \frac{1}{2^n} \cdot \frac{1}{2^n} \varepsilon = \varepsilon \quad \forall \varepsilon > 0$

□
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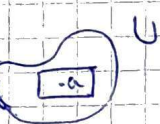
$\therefore \eta(R) = 0$

Teo Cauchy - Goursat ++ Sea $f: U \subseteq \mathbb{C} \rightarrow \mathbb{C}$, U ab, cont. y holomorfe en $U \setminus \{a\}$
 $\Rightarrow \int_{\partial R} f(z) dz = 0$ (R rectángulo)

D/  Si R es tal que $a \notin R$ ✓ Uso la versión anterior

 Si $a \in \partial R$, tomo $R_n \subsetneq R$ tq $a \notin R_n$ tal que R_n se agrandan a R
 $(\alpha - 1/n, \beta) \Rightarrow \int_{\partial R_n} f(z) dz = 0 \quad \forall n \in \mathbb{N}$

Però $\lim_{n \rightarrow \infty} \int_{\partial R_n} f(z) dz = \int_{\partial R} f(z) dz$ $\left. \begin{array}{l} \int_{\partial R} f(z) dz = 0 \\ f \text{ cont. en } R \text{ (compacto)} \end{array} \right\}$

 Si $a \in R^\circ$ parto el rectángulo justo por donde está a y quedan dos rect. que tienen
 $R_1, a \in R_2$ $a \in \partial R_1$ y $a \in \partial R_2$

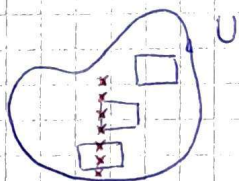
$\int_{\partial R} f(z) dz = \int_{\partial R_1} f(z) dz + \int_{\partial R_2} f(z) dz = 0$

□

¿Y si tengo varios a? $\{a_i\}_{i \in \mathbb{Z}}$, $f: U \rightarrow \mathbb{C}$ cont.

$f: U \setminus \{a_i\}_{i \in \mathbb{Z}} \rightarrow \mathbb{C}$ holomorfe

$a_i \in$ segmento vertical u horizontal



Y pase lo mismo que antes pero cede caso

$g: [0, 2\pi] \rightarrow \mathbb{R}_{>0}$
 $\Rightarrow$ el $\ln$ no se rompe

Teo Fórmula integral de Cauchy en el disco

Sea f continua en $\overline{D(z_0, r)}$, holomorfa en $D(z_0, r)$. Entonces $\forall a \in D(z_0, r)$

$$f(a) = \frac{1}{2\pi i} \int_{\partial D(z_0, r)} \frac{f(z)}{z-a} dz$$

Obs: los valores de f en $\partial D(z_0, r)$ determinan los valores de f adentro de $D(z_0, r)$

D/ Sea $g(z) = \begin{cases} \frac{f(z)-f(a)}{z-a} & \text{si } z \neq a \\ f'(a) & \text{si } z = a \end{cases}$

$\Rightarrow g$ es continua en $\overline{D(z_0, r)}$ y es holomorfa en $D(z_0, r) \setminus \{a\}$

luego, $\int_{\partial D(z_0, r-\epsilon)} g(z) dz = 0 \quad \forall \epsilon > 0$ suf. pequeño tq $a \notin \partial D(z_0, r-\epsilon)$ ← por C-G
 $\uparrow$
 $a \in D(z_0, r-\epsilon)$

$$0 = \int_{\partial D(z_0, r-\epsilon)} g(z) dz = \int_{\partial D(z_0, r-\epsilon)} \frac{f(z)-f(a)}{z-a} dz = \int_{\partial D(z_0, r-\epsilon)} \frac{f(z)}{z-a} dz - f(a) \int_{\partial D(z_0, r-\epsilon)} \frac{1}{z-a} dz$$

$$\Rightarrow f(a) 2\pi i = \int_{\partial D(z_0, r-\epsilon)} \frac{f(z)}{z-a} dz$$

Use conv. unif. pues $f|_{\overline{D(z_0, r-\epsilon)}}$ unif. cont. y como $\lim_{\epsilon \rightarrow 0} \int_{\partial D(z_0, r-\epsilon)} \frac{1}{z-a} dz = 2\pi i$ ⊗

$$\Rightarrow f(a) = \frac{1}{2\pi i} \int_{\partial D(z_0, r)} \frac{f(z)}{z-a} dz$$

⊗ Desarrollo

Como f cont. en $\overline{D(z_0, r)}$ compacto $\Rightarrow f$ unif. cont.

Dado $\eta > 0$, $\exists \delta > 0$ tq $|f(z)-f(w)| < \eta$ si $|z-w| < \delta \quad \forall z, w \in \overline{D(z_0, r)}$

$\gamma(t) = re^{it}$, $\hat{\gamma}(t) = (r-\epsilon)e^{it}$ con $t \in [0, 2\pi]$.

$$\begin{aligned} & \left| \int_{\partial D(z_0, r)} \frac{f(z)}{z-a} dz - \int_{\partial D(z_0, r-\epsilon)} \frac{f(z)}{z-a} dz \right| = \left| \int_0^{2\pi} \left(\frac{f(\gamma(t))}{\gamma(t)-a} - \frac{r-\epsilon}{r} \frac{f(\hat{\gamma}(t))}{\hat{\gamma}(t)-a} \right) \gamma'(t) dt \right| \\ &= \left| \int_0^{2\pi} \frac{f(re^{it})}{re^{it}-a} - \frac{r-\epsilon}{r} \frac{f((r-\epsilon)e^{it})}{(r-\epsilon)e^{it}-a} r e^{it} i dt \right| \leq \quad \text{Ejercicio} \\ &\leq \int_0^{2\pi} \left| \frac{f(re^{it})}{re^{it}-a} - \frac{r-\epsilon}{r} \frac{f((r-\epsilon)e^{it})}{(r-\epsilon)e^{it}-a} \right| \cdot \underbrace{|r e^{it} i|}_{\leq C} dt \\ &\quad \xrightarrow{\epsilon \rightarrow 0} 0 \text{ pues } \end{aligned}$$